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ORIGINAL ARTICLE

Optimization of ultrasound-assisted deep eutectic solvent extraction of betanin and its application in chitosan-based biofilm

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Abstract

A red dragon fruit peel (DFP) contributes to 30–35% of total fruit weight and is disposed of as waste by the fruit processing industries, despite the fact that it contains bioactive compounds with nutritional properties. Ultrasound-assisted deep eutectic solvent (DES) extraction of DFP was performed for the extraction of betanin. From the parametric study, ChCl–ethylene glycol was chosen for further optimization. Extraction parameters were screened using the Plackett–Burman design. Box–Behnken design was applied to get the optimum condition of 40% cycle time, 40% water concentration, and 5 min extraction time to obtain total betanin content of 10.2 mg/10 g dry wt. DFP was valorized to chitosan-based antioxidant films. % DPPH, and % ABTS radical scavenging activity of extract incorporated film was better than control film. Thermogravimetric analysis (TGA) confirmed biofilms could be used safely at food temperature (up to 145 °C). FTIR analysis confirmed the functional group of extract in the chitosan films. With the addition of DES extract, the water vapor permeability and water solubility increased to 3.59×10^{-10} g m^{-1} s^{-1} Pa^{-1} and 47.03%, respectively. The thickness of the film also increased because of the enlargement of space between polymer chains. The elasticity of the films increased by 192.40%, indicating that the DES extract was successfully incorporated as a plasticizer. The tensile strength decreased to 12.46 MPa because of the interaction of DES with the –OH group of chitosan, which disrupts polymer chain bonds. Therefore, bio-refining of waste peel can be advantageous for managing waste disposal and reducing plastic pollution.

Keywords Bio-active film · Chitosan · Deep eutectic solvent · Plasticizer · Natural color extraction

Statement of novelty
The red dragon fruit peels have a waste disposal problem. However, peels are a good and novel source of betanin pigment. The betanin pigments are considered to possess excellent antioxidant and antibacterial properties. Therefore, the betanin was extracted using novel, green, and non-toxic deep eutectic solvents (DES). Also, whenever the extraction is done using DES, the problem of recovery of bioactive compounds takes place, which is a time and energy-consuming process. To counter this problem, a green solution is proposed. The DES extract is directly used as a plasticizer in the fabrication of chitosan-based biofilm. The present work promotes the idea of circular economy by recycling fruit waste into the use of biodegradable film. It reduces the plastic waste problem and encourages value addition to the waste. This is the first study in which DES was used to extract betanin from DFP. Also, this is the first work in which ChCl–ethylene glycol was incorporated into chitosan film.

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1 Introduction

Dragon fruit, called as pitayha or pitaya, is a good source of betalain. Due to its appealing color, nutritional value, and pleasant taste, it is very popular in India [1]. The annual cultivation of dragon fruit is more than 150 tons. It can be classified as white, red, and yellow dragon fruit based on the color of the dragon fruit pulp [2]. In the last few years, red dragon fruit (Hylocereus monacanthus) containing an excellent amount of betalains has attracted consumer interest. Betalains are water-soluble pigments and are reported to have antimicrobial capacity, anticancer properties, anti-inflammation, and antioxidant activity [3]. Recent studies showed that dragon fruit extract has a positive effect on controlling body weight and regulating gut microbiota [4]. Dragon fruit processing industries and their commercial consumption result in vast amounts of solid waste in the form of dragon fruit peels.

Moreover, the weight of peels contributes to 30–35% weight of the whole fruit. As its peels contain a good amount

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of betalain, its direct disposal in the environment results in problems like environmental deterioration and loss of betalain [5]. Therefore, for extraction purposes, its peels can be used as an excellent and novel resource of betalain.

Deep eutectic solvent (DES) usually describes a system wherein two compounds, namely, hydrogen bond donor (HBD) and hydrogen bond acceptor (HBA), interact with each other to form a new solvent with interesting properties like low vapor pressure, easy preparation process, low cost, low toxicity, and highly biodegradable [6]. Because of the creation of a hydrogen bond between the DES component and extracted compounds, DES is generally regarded as a suitable solvent for extraction purposes. The formation of bonds results in a high yield and good stability of extract [7]. Moreover, the disadvantage of using DES in extraction is its high kinematic viscosity (200–500 mm^2/s), resulting in difficulty in diffusivity. Higher temperature and water addition are reported to reduce viscosity. Along with that, novel extraction techniques like stirring, microwave-assisted, and ultrasound-assisted extraction have synergized effects on extraction [8].

Chitosan is biodegradable, non-toxic, polycationic copolymer which is obtained from the deacetylation of chitin [9]. Pure chitosan films are rigid and brittle. Therefore, DES is added as a plasticizer [10]. Plasticizer promotes polymer chain mobility and decreases intermolecular forces, improving the film's extensibility and flexibility [11]. From an environmental point of view, chitosan films are more attractive and desirable because chitosan can be manufactured from shell waste. Therefore, the chitosan packaging will obtain material from renewable sources and reuse the waste material [12].

The present study's hypothesis aims to employ DES as an efficient extraction medium for betanin. For this purpose, ultrasound-assisted extraction of total betanin was explored using different DES, and extraction conditions were optimized using the Box–Behnken design (BBD) approach. After extraction using DES, problem of the recovery of bioactive compounds arises. The DES extract was used as a plasticizer to prepare the chitosan-based bioactive film to overcome this problem. Bioactive film can have further applications in the food packaging industry. Betanin compounds extracted from red dragon fruit peels can increase the antioxidant property of biofilm.

2 Material and methods

2.1 Raw material and chemicals

Red dragon fruits were procured from a local market in Surat, Gujarat. Medium molecular weight chitosan (90% deacetylation degree, CAS No. 9012-76-4), standard betanin (CAS No. 7659-95-2), HPLC grade acetonitrile (CAS No. 75-05-08), water (CAS No. 7732-18-5), and trifluoroacetic acid (CAS No. 76-05-1) were purchased from Sigma-Aldrich Chemical Co. Ltd., India. Choline chloride (98%, CAS No. 67-48-1), glycerol (98%, CAS No. 56-81-5), ethylene glycol (99%, CAS No. 107-21-1), calcium chloride fused (98%, CAS No. 10043-52-4), and urea (99%, CAS No. 57-13-6) were procured from Finar Limited, Ahmedabad.

2.2 Dragon fruit peel powder preparation

Peels from each dragon fruit were removed by knife and cleaned with water. The peels were cut into 3 × 4 cm small size and were put into a deep freezer overnight. The pieces of dragon fruit peels were freeze-dried inside a vacuum chamber for 16 h at − 40 °C. After freeze-drying, it was subjected to a high-speed mixer (Blender 7012S; Waring, Torrington, India). The resulting dragon fruit powder was passed through a 60 mesh size sieve to get less than 250 µ particles. The prepared powder was then kept in a polyethylene bag and kept in a desiccator until further use.

2.3 DES preparation

In this study, ChCl: urea (DES-1), ChCl: glycerol (DES-2), and ChCl: ethylene glycol (DES-3) were prepared, in which HBD and HBA were subjected to heating and stirring at 80 °C using an automatic heater and stirrer (Remi 5 MLH plus, India) until the homogenous transparent clear liquid was formed. The prepared DES was then stored at room temperature.

2.4 Ultrasound-assisted extraction

Sonication was carried out using an ultrasonic processor (Model-UP200Ht, frequency 26 kHz, power 200 W), which was equipped with a digitally controlled system for temperature, time, and power control. A piezoelectric convertor was attached, and a titanium alloy probe (7 mm diameter) was used in the batch system. The solid-to-liquid ratio of 0.2 g dragon fruit dried powder per mL of solvent was used for this extraction, which was performed according to the experimental design. The samples were diluted, centrifuged, and sent for analysis of betanin content. The effect of different molar ratios, as well as water concentration, was investigated for all DES used. The extraction process parameters like pulse ratio, amplitude, solid–liquid ratio, and time were also optimized to achieve a high total betanin yield.

2.5 Plackett–Burman design

In this work, the Placket–Burman design (PBD) was applied to find the significant factors [13]. The factors evaluated

were water concentration, cycle time, amplitude, solid-liquid ratio, extraction time, and molar ratio. The design included 12 runs of various combinations of six independent factors with their corresponding high (+) and low (−) levels. Before PBD experiments, the high and low levels were chosen based on a parametric study shown in supplementary data (Figure S3). Total betanins were taken as the response. All experiments were performed in duplicates, and their average was taken into account. PBD assumes no interactions between the factors; each factor is independent and can be described by the first-order model:

$$Y = \beta_0 + \sum \beta i + X_i \quad (1)$$

where Y is the predicted response, β_0 is the model intercept, β_i is regression co-efficient, and X_i is independent parameter.

2.6 Optimization by Box–Behnken design

Based on the parameters screened by PBD, the extraction parameters were further optimized by BBD. Cycle time, water concentration, and extraction time were optimized in the range of 30–50%, 20–40%, and 5–10 min, respectively. Each experiment was performed in duplicates, and its average value was taken into account. The effect of these parameters was analyzed using RSM. Fifteen runs in a random manner were performed, and three runs were performed at the center point. The results were fitted by response surface regression:

$$Y = \beta_0 + \sum \beta_i X_i + \sum \beta_{ii} X_i^2 + \sum \beta_{ij} X_i X_j \quad (2)$$

where Y is the predicted response, X_i and X_j are coded independent variables, and β_0, β_i, β_{ii} and β_{ij} are regression co-efficient of intercept, linear, quadratic, and interaction terms, respectively. Design expert software was utilized for regression, experimental design, and graphical analysis of data.

2.7 HPLC analysis

Quantification of the prominent peak observed in the extract was carried out using the betanin standard. HPLC analysis with modifications according to a previous study [14] was done using HPLC system (Shimadzu, model LC-2030 3D) and a PDA detector with an analytical column reverse phase C18 column (250 mm × 4.6 mm id) with a particle size of 5 µm. The isocratic method used the mixture of 0.1% trifluoroacetic acid in 90% water and 10% acetonitrile as the mobile phase. The detection was made at 536 nm, and 10 µL of the sample was allowed to elute through the system for 20 min at 0.5 mL/min of flow rate. Also, the column temperature was kept at 30 °C. The samples were filtered through a 0.45 µ membrane filter before HPLC analysis. The HPLC chromatograms of standard betanin and DES-extract are shown in Figures S1 and S2, respectively. From the chromatogram of standard betanin, two compounds, namely, betanin and isobetanin, were identified. However, the extract contains four betacyanin compounds. As suggested by Roriz et al. [15], the prominent peak of betanin (40%) was found to be at a higher concentration. In comparison, the minor peak of isobetanin was detected at a lower concentration (less than 6% of the total content), and the optimization was performed for total betanin (sum of betanin and isobetanin).

2.8 Biofilm preparation

The method of fabrication of chitosan films was modified from a previous study in which the first step is adding 2 wt% chitosan in 49 mL of 2% (w/v) acetic acid solution [16]. The resulting mixture was kept under overnight stirring. Subsequently, 30 wt% (relative to the mass of chitosan) DES extract was incorporated into the mixture and subjected to stirring for 30 min. The viscous mixture was subjected to sonication to remove dissolved gases and air bubbles. Then, 25 mL of the mixture was poured into a petri dish of a defined surface area and dried in an oven for 24 h at 50 °C. After that, they were kept in a desiccator until further use.

2.9 DPPH antioxidant activity

The antioxidant capacity of chitosan films and their enhancement by the addition of DES extract was measured by DPPH radical scavenging activity. Samples were prepared according to previously published literature, in which 500 µL supernatant was mixed with 2 mL methanolic DPPH (0.06 mM) solution. The resulting mixture was then vigorously shaken and kept in the darkroom for 30 min. The absorbance was measured at 517 nm using a microplate reader (SPECTROstar nano, BMG labtech). This process was carried out in duplicate. The following equation was used to calculate %DPPH inhibition [17].

$$\%\text{DPPH inhibition} = \left(1 - \frac{A_{\text{Sample}}}{A_{\text{Control}}}\right) * 100 \quad (3)$$

where A_{sample} is the absorbance of the sample and A_{control} is the absorbance of a blank.

2.10 ABTS$^+$ radical cation method

300 µL of ABTS stock solution and 100 µL supernatant were mixed for 30 min. Then, absorbance was measured at 732 nm using a microplate reader (SPECTROstar nano, BMG labtech). This process was carried out in duplicate. The following equation was used to calculate %ABTS inhibition [18].

$$\%\text{ABTS inhibition} = \left(1 - \frac{A_{\text{Sample}}}{A_{\text{Control}}}\right) * 100 \quad (4)$$

2.11 Antioxidant capacity of films

Samples were prepared according to a previous study, in which 2 mL methanol was mixed with 100 mg of film and subjected to stirring for 3 h at room temperature. The sample was homogenized in a vortex, and the supernatant was taken for analysis [17]. %DPPH inhibition/100 mg of film and %ABTS inhibition/100 mg of the film were measured for films.

2.12 FTIR analysis

Vibrational spectra of studied samples were recorded using a Shimadzu (IRAffinity-1 s) spectrometer within the wavelength range from 4000 to 500 cm^{-1} (45 scans, spectral resolution of 4 cm^{-1}). The spectra of all studied samples were obtained at room temperature.

2.13 Thermogravimetric analysis (TGA)

For the investigation of the thermal stability of biofilms, a thermal gravimetric analyzer (NETZSCH TG 209F1 libra) was used under an inert atmosphere with approximately 6–12 mg sample quantity with a ramp-up rate of 20 °C and a temperature range from 30 to 700 °C.

2.14 Mechanical behavior of film

The tensile strength (TS) is referred to as the maximum load that film can bear without breaking apart, and the distance at burst (DB) is referred to as the total elongation of a film before rupture [19]. It was measured using the Instron-5967 Universal testing machine. The test was performed in tensile mode at a rate of 10 mm min^{-1} with a film specimen size of 10 × 1 cm (L × W). The film thickness was measured using the handheld baker dial thickness gauge (0.01–10 mm). At four different locations, it was measured, and its average was considered.

2.15 Water vapor permeability

Using the method described by Galvis-Sanchez et al. [20], the water vapor permeability was calculated. The dried calcium chloride with a predefined weight was placed in a glass container of 5 cm diameter. The top of the container was tightly sealed with film. The containers were placed in a desiccator over saturated sodium chloride solution at room temperature (75% relative humidity). Weight gain by the container is measured every 1 h till it reaches a constant value. Three tests were performed, and the average was taken into account.

The water vapor permeability (WVP) was calculated using the following equation:

$$\text{WVP}\left(\text{g} \cdot \text{m}^{-1} \cdot \text{s}^{-1} \cdot \text{Pa}^{-1}\right) = \frac{w \times x}{A \times t \times \Delta p} \quad (5)$$

where w is the weight gained by the container (g), x is the thickness of the film (m), A is the area of the exposed film (m^2), t is time(s), and Δp is the partial pressure of water (Pa).

2.16 Water solubility (WS)

Biofilms were cut into 25 mm × 25 mm pieces and dried in a 100 °C oven for 24 h, and their initial weight (M_i) was measured. At room temperature, the film sample was immersed in 50 mL of water for 24 hour period of time. The sample was taken out of the water, subjected to drying in an oven at 100 °C for 24 h, and weighed to determine its final weight (M_f).

$$\text{WS}(\%) = \frac{M_i - M_f}{M_i} \times 100 \quad (6)$$

3 Results and discussion

3.1 Plackett–Burman design(PBD) for screening of solvent

Six parameters, namely, solid–liquid ratio, extraction time, water %, cycle time, amplitude, and molar ratio of deep eutectic solvent, were considered for the study. Plackett–Burman design (PBD) was utilized to screen the six parameters mentioned above. Li et al. [13] also used a combination of PBD followed by Box–Behnken design (BBD) to extract anthocyanins from blackcurrent marc. The authors screened six parameters using PBD. Then, three significant parameters were optimized further using BBD. Chen et al. [21] also used a similar approach to extract samara oil from *Acer saccharum.* Seven parameters for efficient extraction were screened using PBD. Three significant factors found using PBD design were further optimized using BBD.

From the parametric study in Figure S3, DES-3 was chosen for further optimization. The PBD design matrix is shown in Table 1. The factors with a high confidence level ($p < 0.05$) are considered significant parameters. As shown in Figure S4, the Pareto chart represents the effect of various factors on the response and checks its statistical significance. A parameter with a t value greater than the t value limit can be considered a significant parameter. As shown in Table 2, the R^2 value of 0.90 for total betanin shows that Pareto chart values are

Table 1 Plackett–Burman experimental design

Run no.	Water concentration (%)	Cycle time (%)	Amplitude (%)	Solid–liquid ratio	Time (min)	Molar ratio	Total betanin (mg/10 g dry weight)
1	40	50	25	0.2 g/40 mL	5	1 : 2	8.75
2	20	50	50	0.2 g/20 mL	10	1 : 2	5.40
3	40	50	25	0.2 g/40 mL	10	1 : 2	8.23
4	40	30	50	0.2 g/20 mL	5	1 : 2	9.83
5	40	30	50	0.2 g/40 mL	5	1 : 5	9.85
6	20	30	25	0.2 g/40 mL	10	1 : 5	8.39
7	40	30	25	0.2 g/20 mL	10	1 : 5	9.95
8	20	30	50	0.2 g/40 mL	10	1 : 2	7.85
9	40	50	50	0.2 g/20 mL	10	1 : 5	7.98
10	20	50	50	0.2 g/40 mL	5	1 : 5	8.57
11	20	50	25	0.2 g/20 mL	5	1 : 5	8.25
12	20	30	25	0.2 g/20 mL	5	1 : 2	8.80

Table 2 Analysis of variance for PBD

Source	DF	Adj SS	Adj MS	*F* value	*p* value
Model	6	14.70	2.45	7.59	0.02
Linear	6	14.70	2.45	7.59	0.02
Water concentration	1	4.47	4.47	13.84	0.01
Cycle time	1	4.68	4.68	14.49	0.01
Amplitude	1	0.70	0.70	2.17	0.20
Solid–liquid ratio	1	0.17	0.17	0.53	0.49
Time	1	3.26	3.26	10.1	0.02
Molar ratio	1	1.41	1.41	4.38	0.09
Error	5	1.61	0.32		
Model summary					
	R-sq	*R*-sq (adj)	*R*-sq (pred)		
	90.10%	78.22%	42.99%		

dependable. Water concentration, cycle time, and extraction time can be considered significant parameters from the Pareto chart. Therefore, they were further optimized in the subsequent investigation, while other parameters were kept constant.

The model equation developed by PBD for total betanin is as follows:

$$\text{Total betanin} = 8.49 + 0.61 \times \text{water concentration} - 0.62 \times \text{cycle time} - 0.24 \times \text{amplitude} + 0.12 \times \text{solid-liquid ratio} - 0.52 \times \text{time} + 0.34 \times \text{molar ratio} \quad (7)$$

3.2 BBD design

Amplitude (50%), solid–liquid ratio (0.2 g/ 20 mL), and molar ratio (1 : 2) were kept constant, while cycle time, water concentration, and time were further optimized in order to increase the total betanin yield. Table 3 represents the experimental design using the Box–Behnken design.

With multiple regressions, the following equation was found to be best suitable:

$$\text{Total betanin} = 8.99 - 0.67 \times A + 0.62 \times B - 0.93 \times C + 0.14 \times AB - 0.65 \times AC + 0.32 \times BC - 4.66 \times A^2 - 0.16 \times B^2 + 0.06 \times C^2 \quad (8)$$

As shown in Table 4, ANOVA was performed to obtain the statistical significance of the quadratic model with $p < 0.05$. The model was found to be significant with $F = 161.31$ and $p < 0.01 \times 10^{-2}$ with high regression co-efficient $R^2 = 0.99$. The value of adjusted R^2 was found to be 0.99, which shows the excellent fitting of the data point into the regression curve. No significant difference was found between actual R^2 and adjusted R^2. Lack of fit (0.03) was found to be not significant. The $R_{pred}{}^2$ was 0.96, which was very near to $R_{adj}{}^2$. The adequate precision was 43.25, which shows the excellent ability of the mathematical model to predict the observation. At optimum conditions, the total betanin yield was found to be 10.20 mg/10 g dry weight, while the predicted yield at the same condition was 10.14 mg/10 g dry weight. We also performed a confirmation test at optimum conditions and obtained 10.06 mg total betanin/10 g dry weight. In the literature, total betacyanin content of 8.4 mg/10 g for Indonesian, 13.1 mg/10 g for Spanish, and 9.3 mg/10 g for Mexican species of red dragon fruit is reported [22].

Different parameters like cycle time, extraction time, and water concentration significantly affected total betanin yield. Their interactive effect on total betanin yield is shown in 3D graph. A 3D graph is a valuable tool for understanding the interactive effect. The interactive effect of water concentration and cycle time on total betanin is shown in Fig. 1a. The total betanin yield is low at low water concentration and high cycle time, while

Table 3 Experimental planning using BBD design

Run	Independent variables			Total betanin yield (mg/10 g dry weight)	
	Cycle time (%)	Water concentration (%)	Time (min)	Experimental	Predicted
1	30	20	7.5	8.59	8.55
2	50	20	7.5	7.00	6.92
3	30	40	7.5	9.43	9.52
4	50	40	7.5	8.41	8.45
5	30	30	5	9.57	9.55
6	50	30	5	9.49	9.51
7	30	30	10	9.01	8.98
8	50	30	10	6.30	6.32
9	40	20	5	9.46	9.53
10	40	40	5	10.20	10.14
11	40	20	10	6.95	7.01
12	40	40	10	8.96	8.90
13	40	30	7.5	8.96	8.99
14	40	30	7.5	8.90	8.99
15	40	30	7.5	9.10	8.99

at 30–40% water concentration and cycle time between 30 and 40%, the total betanin yield is higher. With the increase in water concentration, DES viscosity was found to be decreasing. This may contribute to the higher diffusion of betanin to the solvent. Figure 1b describes the interaction between cycle time and extraction time and its effect on total betanin yield. It is clear that at a very high cycle time of 45–50% and extraction time of 8–10 min, the betanin yield decreases significantly. During the experiment, it was observed that a very high cycle time for a prolonged extraction time resulted in a temperature rise above 80 °C. High ultrasonic power increases extraction yield by improving the acoustic cavitation effect. But, very high ultrasonic power reduces the yield by favoring agitation over cavitation [23]. In the literature, a decrease in bioactive compound yield was observed at prolonged extraction time and cycle time [24]. High temperature favors the extraction yield by increasing the mass transfer of betanin. But, very high temperatures (above 80 °C) may result in the degradation of betanin by carboxylation, isomerization, and deglycosilation reactions. This decreases the betanin yield [25]. However, the cycle time of 30–40% and extraction time of 5–7 min results in a good extraction yield. Figure 1c describes the interaction between the water concentration and time and its effect on total betanin yield. Low water concentration results in high viscosity of DES. Also, a lower value of extraction time results in the decreased mass transfer of betanin. To achieve the maximum betanin yield, the water concentration should be 30–40%, and extraction time should be 5–8 minutes.

3.3 Biofilm preparation

After the DES extraction, the problem of recovery arises. The DES extract was directly used as a plasticizer to fabricate biofilm to solve this problem. This chitosan-based biofilm can be used as a packing material. This can be an environmentally friendly alternative to plastics. The biofilm prepared using the previously mentioned technique is shown in Fig. 2.

Table 4 Analysis of variance of the quadratic model for BBD design

Source	Sum of squares	Degree of freedom	Mean square	*F* value	*p* value	
Model	16.95	9	1.88	161.31	$< 0.01 \times 10^{-2}$	Significant
A: pulse ratio	3.65	1	3.65	312.46	$< 0.01 \times 10^{-2}$	
B: water concentration	3.13	1	3.13	268.33	$< 0.01 \times 10^{-2}$	
C: time	7.03	1	7.03	602.57	$< 0.01 \times 10^{-2}$	
AB	0.08	1	0.08	7.01	0.04	
AC	1.73	1	1.73	148.14	$< 0.01 \times 10^{-2}$	
BC	0.40	1	0.40	35.10	0.2×10^{-2}	
A^2	0.80	1	0.80	68.69	0.04×10^{-2}	
B^2	0.09	1	0.09	8.53	0.03	
C^2	0.01	1	0.01	1.55	0.26	
Residual	0.05	5	0.01			
Lack of Fit	0.03	3	0.01	1.15	0.49	Not significant
R^2	0.99	R_{adj}^2	0.99	R_{pred}^2	0.96	
Adequate precision	43.25					

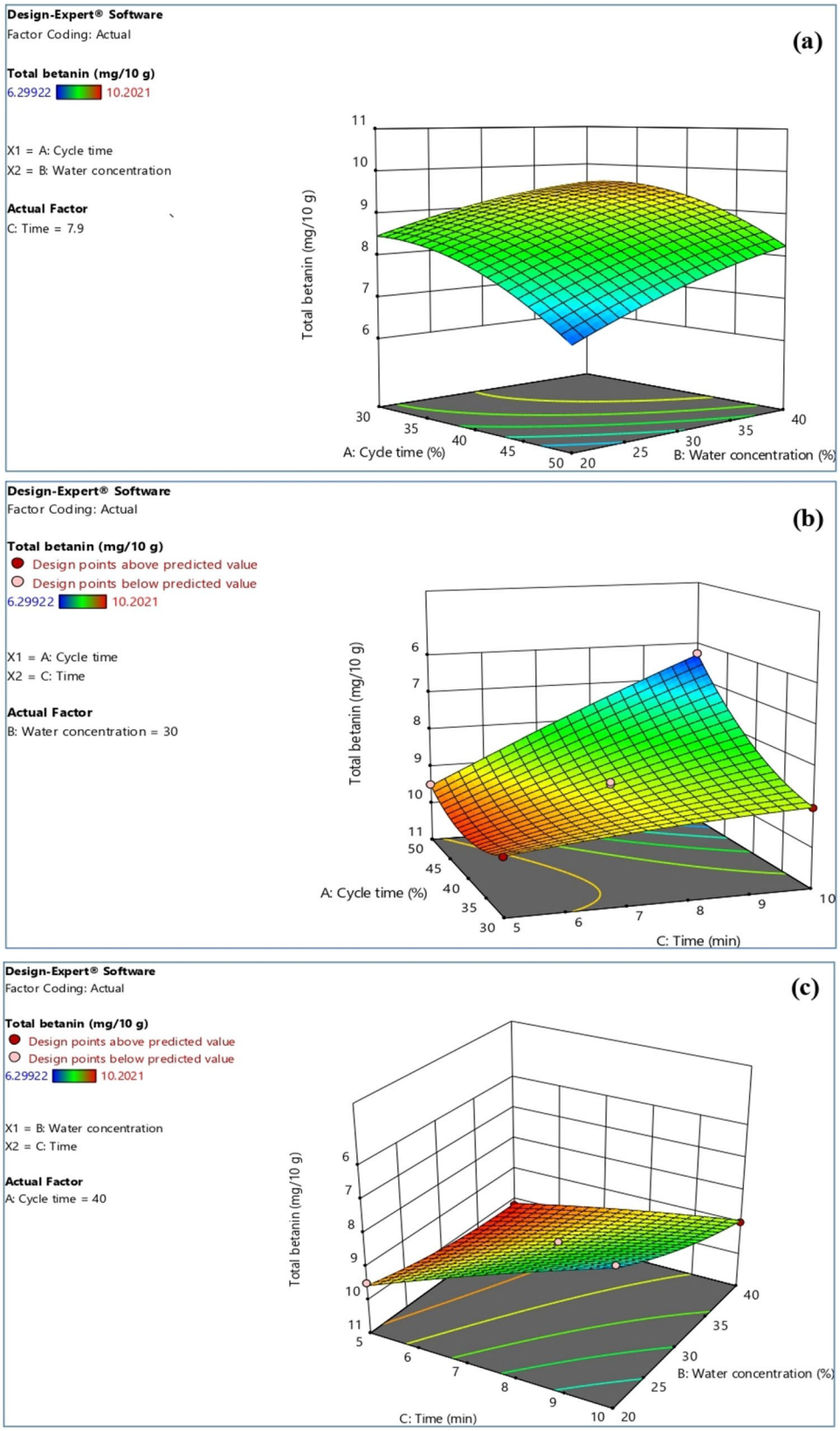


Fig. 1 Interactive plot for (**a**) water concentration and cycle time, (**b**) cycle time and extraction time, and (**c**) water concentration and extraction time

3.4 Antioxidant activity

As shown in Table 5, the antioxidant activity of extract as well as films was studied by DPPH and ABTS radicals. All the extracts showed high antioxidant activity. Moreover, the %DPPH scavenging activity and %ABTS scavenging activity were in the range of 62.05 to 67.85% and 77.73 to 83.19%. Betacyanin compounds like betanin, betanidin, and phyllocactin have strong antioxidant activity in red dragon fruit. Also, betacyanin compounds can be used as a natural defense against oxidative stress [26]. Betalain-related metabolites were reported to be the main contributors to antioxidant activity in pitayas [27]. In literature, %ABTS scavenging activity and %DPPH scavenging activity of different dragon fruit species were in the range of 61.80 to 89.80% and 55.6 to 81.2% [28].

Oxidation is the primary reason why the degradation of food products takes place. The oxygen reacts with food products containing pigments and fat. This can result in the creation of unwanted low molecular weight compounds responsible for unpleasant odor and taste and the destruction of vital nutrients like vitamins [29]. The oxidized product also tends to change the color of food, making it unappealing from the consumer's point of view [16]. %DPPH scavenging activity of different films showed no significant difference. All films showed similar scavenging activity with %DPPH inhibition per 100 mg of film between 17.61 ± 1.41 and 20.98 ± 0.91%. The chitosan film without any extract also showed little DPPH scavenging activity, similar to the study by [30, 31]. For all the films, the %ABTS inhibition per 100 mg of the film was in the range of 32.75 ± 0.56 to 43.80 ± 0.60%. The control film showed the lowest %ABTS inhibition, while the DES-3 reported the highest %ABTS inhibition.

The marginal increase in antioxidant activity of the DES extract (30 wt%) incorporated film was observed. This could be because of the precursor. Velásquez et al. [18] performed extraction of anthocyanin from the Chilean Luma chequen (Molina) A. Gray berry using different DES and further incorporated anthocyanin-rich DES extract (30 wt%). The author observed % DPPH inhibition between 7.35 and 10.81% and %ABTS inhibition between 21.72 and 30.50% for different DES extract-based films. This data shows no significant increase in the DPPH inhibition (%) and ABTS inhibition (%) as compared to the control film. It seems like anthocyanin, and betacyanin-rich DES-extract has limited effect on improving the antioxidant activity of biofilm. However, increasing the quantity (wt%) of DES extract can improve the antioxidant property.

Table 5 Antioxidant activity of different extracts and films

	%DPPH scavenging activity	%ABTS scavenging activity
DES-1 extract	67.85 ± 0.59	83.19 ± 0.93
DES-2 extract	68.82 ± 1.95	77.64 ± 0.20
DES-3 extract	68.91 ± 0.94	78.64 ± 0.74
Water extract	62.05 ± 0.9	77.73 ± 0.04
Control film	17.61 ± 1.41	32.75 ± 0.56
DES-1 extract film	20.98 ± 0.91	41.87 ± 1.04
DES-2 extract film	19.12 ± 0.17	43.40 ± 1.32
DES-3 extract film	20.41 ± 0.50	43.80 ± 0.60

3.5 TGA

To determine the thermal stability of the biofilm, the TGA analysis is considered the most reliable method. As shown in Fig. 3, TGA of pure chitosan film (control film), DES-1 extract film, DES-2 extract film, and DES-3 extract film was performed. Three stages of thermal degradation were observed.

Up to 145.35 °C, the mass loss was due to the evaporation of physically absorbed water. As shown in Table S1, the control film (12.39 wt%) and DES-3 extract film (12.83 wt%) lost the highest mass %, which indicates that under the external environmental conditions, the DES-1 extract film and DES-2 extract film are less hydrophilic than the control film. In the second thermal degradation stage, the temperature range was from 145.35 to 217.35 °C. The lowest weight loss was observed for the control film (8.1 wt%); however, plasticized films showed a high mass loss. For DES-1 extract film, this mass loss could be attributed to the breakdown of urea into biuret, cyanin acid, and ammelide at a temperature range of 120–216 °C. While for DES-2 film, this mass loss could be because of the evaporation of glycerol. For DES-3 film, evaporation of ethylene glycol takes place between 80

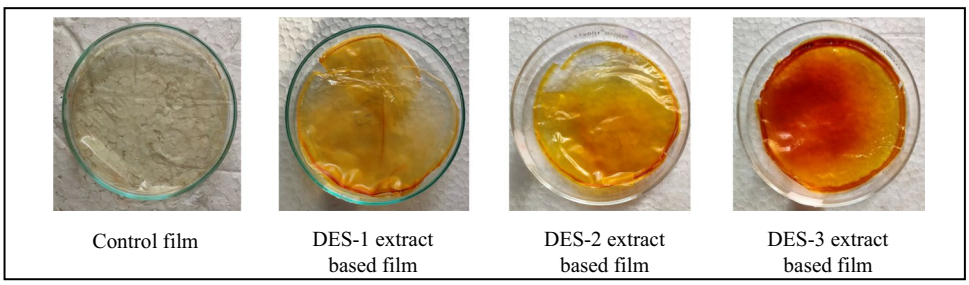


Fig. 2 Biofilm prepared from chitosan with 30% DES extract plasticizer

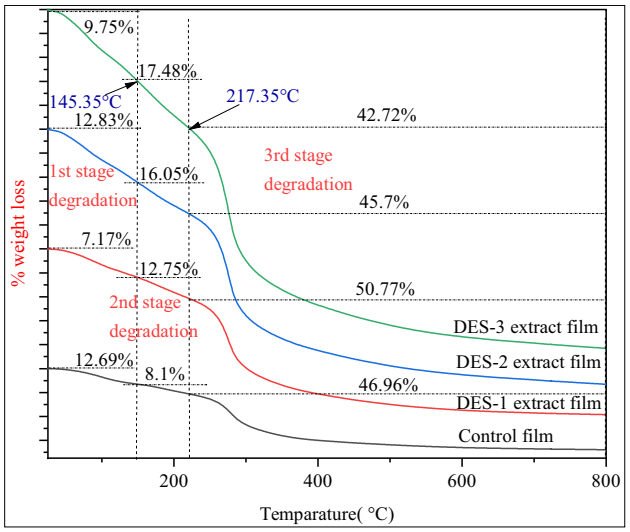


Fig. 3 TGA analysis of control film and DES extract-based film

and 242 °C [32]. The third stage of degradation takes place between the temperature range of 217.35 to 799.35 °C. The highest weight loss was observed for DES-1 extract film, with 50.77% mass loss, while DES-3 showed the lowest weight loss of 42.72%. In DES extract films, the mass loss could be attributed to the decomposition of ChCl between 281 and 308 °C. Boiling of ethylene glycol reaches and exceeds between 242 and 308 °C. Also, glycerol vaporization takes place at 289 °C [32]. Some authors have shown that weight loss due to chitosan takes place in two stages. The 1st stage starts at 60 °C with 10% weight loss. 2nd stage of degradation occurs between 240 and 380 °C with 41.4% mass loss [33]. From the above findings, it can be safely assumed that biofilms can be used as a packaging material with a food temperature of less than 145.35 °C. However, the evaporation of water may cause some changes in its properties based on the content of water present [9].

3.6 FTIR analysis

FTIR of pure chitosan, all DES, their extract, pure DES film, and their corresponding DES extract film is presented in Fig. 4. For DES-3, wavenumber 3375 cm^{-1} refers to O-H stretching of hydroxyl groups. Vibrational bands at 2983 and 2885 cm^{-1} represent C-H stretching of an sp^3 hybridized bending, 1467 cm^{-1} to CH_2 bending of the alkyl group, and 1257 cm^{-1} to C-O-H bending, 1159 cm^{-1} C-O stretching, and 1043 cm^{-1} to C-C-O asymmetric stretching. For the vibrational spectrum of DES-1, wavenumber 3394 cm^{-1} represents symmetric NH_2 stretching, 1448 cm^{-1} to the CH_2 bending of the alkyl group, and 1624 cm^{-1} to the functional group of urea. Same vibrational bands were observed for DES-2 and DES-3 and their corresponding extracts because of molecular similarity between ethylene glycol and glycerol. Similar results were found in the literature as well [32]. The distinct peak at 3395, 2978, 2154, 1650, 1467, 1421, 1121, and 1014 cm^{-1} represents the betalain group present in the extract [34]. All films showed vibrational bands in the range of 3367–3302 cm^{-1} and 2954–2889 cm^{-1} indicating the presence of the N-H group. DES extract films showed the presence of N=C=S stretching and C=C stretching at the wavenumber range of 2117–2096 cm^{-1} and 1710–1625 cm^{-1}, while DES films and control films lacked this peak. Control film and DES films showed the presence of the C=N group at the 1710–1641 cm^{-1} vibrational range. All films showed the presence of N-O stretching and O-H bending in the range of 1550–1543 cm^{-1} and 1409–1415 cm^{-1}. DES films and DES extract films showed strong C-O stretching (1244–1230 cm^{-1}). All films showed the presence of C-O stretching and C-C-O bands at 1155 cm^{-1} and 925 cm^{-1}. Pure chitosan showed a peak at 1031 cm^{-1} and 1068 cm^{-1} representing C-O-C and C-O-H vibrations. Similar results were observed in the literature as well [35].

3.7 Mechanical properties of the film

The average thickness (μm) measured for the chitosan films with different DES extracts is presented in Table 6. The thickness value for pure chitosan film (97.50 μm) was lower than plasticized films. The thickness of plasticized films was in the range of 112.50 to 117.50 μm. The literature also observed the thickness of chitosan films in the range of 94 to 120 μm [17]. The increase in thickness could be because of plasticizer use, which causes an enlargement in space between the polymer chains [19].

Mechanical properties are related to measuring the film's ability to protect the integrity of the foods [17]. Table 6 represents the average tensile strength (TS) and the average distance of elongation (DE) of control film and films prepared using DES extract. It can be observed that films prepared using DES extract showed more elasticity than control film but low resistance to breaking apart. The TS of DES-1 extract, DES-2 extract, and DES-3 extract films was 12.46 ± 1.59, 14.12 ± 1.91, and 14.17 ± 2.73 MPa, respectively, and were found to be lower as compared to the control film. Similar values of TS for the control film were observed in the literature [36, 37]. Literature reported TS of 17 MPa for chitosan film with 33 wt% glycerol as plasticizer [38]. The decrease in tensile strength with the addition of DES can be explained by the interaction of DES with the –OH group of chitosan, which disrupts polymer chain bonds. The chitosan-chitosan interaction is responsible for the mechanical integrity of the chitosan-based biofilm. This interaction forms a grid-like structure. When the DES is added, chitosan-DES interaction disrupts this grid-like structure. This may be

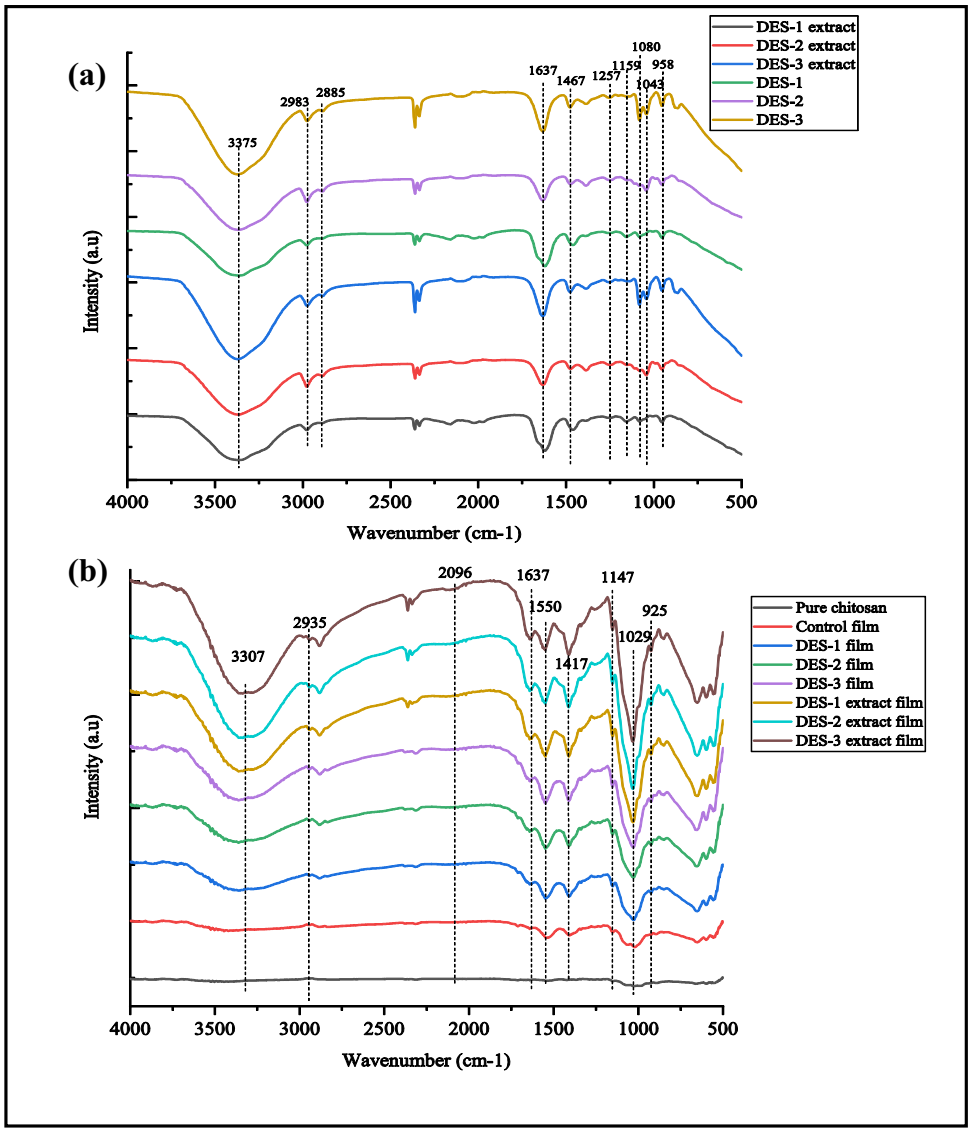


Fig. 4 FTIR of (a) different DES and their extracts and (b) prepared films

Table 6 Mechanical properties and water properties of the film

Films	Thickness (μm)	TS (MPa)	DE (mm)	WVP($\times 10^{-10}$ g m^{-1} s^{-1} Pa^{-1})	WS (%)
Control film	97.50	25.96 ± 5.49	0.79 ± 0.12	3.01 ± 1.33	24.57 ± 3.13
DES-1 extract film	117.50	12.46 ± 1.59	2.19 ± 0.24	3.59 ± 1.08	47.03 ± 6.76
DES-2 extract film	117.50	14.12 ± 1.91	2.31 ± 0.67	3.51 ± 1.01	46.85 ± 5.18
DES-3 extract film	112.50	14.17 ± 2.73	1.99 ± 0.47	3.18 ± 1.48	45.62 ± 1.73

explained by a lower molecular weight of DES as compared to the chitosan. This results in a decrease in tensile strength with the addition of DES. Therefore decrease in tensile strength with the addition of DES is observed [39]. The DE of the control film was 0.79 ± 0.12 mm. The DE was enhanced by 2.19 ± 0.24 mm, 2.31 ± 0.67 mm, and 1.99 ± 0.47 mm for DES-1, DES-2, and DES-3 extract film, respectively.

The elasticity of chitosan-based film is expected to rise with the addition of DES. This could be due to the improvement of free space between the polymer chains [19]. For chitosan film based on polyvinyl alcohol and bacterial cellulose, a DE of 0.64 to 5.12 mm was reported [40]. Other parameters also play a part in increasing the elasticity of the film. The incorporation of the DES extract makes the film more hydrophilic. In other words,

it improves the water content of the film. High water content also increases the film elasticity as water can act as a plasticizer [17]. Water, with its low molecular weight, increases the molecular mobility of amorphous polymers because of higher free volume [41]. Kyriakidou et al. [10] also observed the effect of water content on film elasticity and found that high water content increased film elasticity significantly. Jakubowska et al. [9] added choline chloride–malic acid as a plasticizer up to 80 wt% and observed 171.3% elongation at break as compared to 3.3% elongation at break for the control film. Polymer chain translation is mainly responsible for the stability of film in the elastic region of deformation. An increase in DES amount was found to be related to an increase in free volume in the polymer matrix, which is essential for polymer chain translation. Cazón et al. [42] also observed plasticizing effect of *Melaleuca alternifolia* essential oil in chitosan film. The author explained this effect by substituting strong polymer interaction with polymer–essential oil interaction in the film matrix.

The mechanical property of DES and DES extract incorporated biofilms was reported in previous literature. Jakubowska et al. [9] developed chitosan-based films with the addition of choline chloride–malic acid. The author found that the addition of DES into films up to 80% wt in chitosan films decreased TS up to 1.5 MPa and increased % elongation at break up to 171.3%. Sokolova et al. [39] also observed that the addition of choline chloride–malonic acid up to 67 wt% results in a decrease in TS and an increase in elasticity. Chandra et al. [35] incorporated astaxanthin-rich choline chloride–lactic acid extract in the chitosan film. The authors observed an increase in TS up to 28.09 MPa and a decrease in elongation at break up to 32.17% with the addition of DES extract, which is in contradiction with our result. The presence of astaxanthin was found to be the reason for this phenomenon. Alves et al. [43] used choline chloride–lactic acid, choline–glycerol, and choline chloride–glycerol–lactic acid for polyphenolic extraction from green tea. Furthermore, the author used all three DES and DES extract as plasticizers up to 30 wt%. A similar trend in mechanical property was observed.

3.8 Water vapor permeability of film (WVP) and water solubility (WS)

The film ability to control how easily the moisture can penetrate and transfer through the hydrophilic portion of the film is known as WVP [44]. The value of WVP differs in other literature because of different molecular weight, degree of deacetylation, types of plasticizer and their concentration, different types of film preparation techniques, use of different types of solvents, and their concentration. Moreover, the method and condition (temperature and relative humidity) of measuring WVP and film thickness also affect the WVP value [9].

For a film to be an excellent packaging material, there is no specified range of WVP. It mainly depends upon the type of product to be packaged. A high moisture level is required for a wet food product (e.g., cheese and meat). On the other hand, a low moisture level is necessary for dry food products like dry pet food and snacks [9].

As shown in table 6, WVP of the control film (3.01×10^{-10} g m^{-1} s^{-1} Pa^{-1}) was the lowest among all other films. WVP of DES-1 extract film (3.59×10^{-10} g m^{-1} s^{-1} Pa^{-1}) was observed to be the highest, while DES-2 extract film (3.51×10^{-10} g m^{-1} s^{-1} Pa^{-1}) and DES-3 extract film (3.18×10^{-10} g m^{-1} s^{-1} Pa^{-1}) showed lower value of WVP. However, the DES extract films generally showed more WVP values than the control film. It can be observed that DES extract used as a plasticizer is hydrophilic and increases the water vapor diffusion path. The higher value of WVP for the DES extract modified film can be explained by the penetration of chitosan–chitosan chains by the DES molecule, which decreases intermolecular bonding and boosts segmental chain mobility. This increases the transfer of water molecules [9]. Similar values of WVP was found in literature as well [20].

Water solubility (WS) is one of the most critical film parameters for its application in food packaging. As shown in Table 6, the WS of the control film is 24.54 ± 3.13%, which is nearer to the published literature [36, 45]. An increase in water solubility with the addition of DES extract can be explained by the hydrophilic nature of DES and total betanin. Almeida et al. [46] also observed an increase in WS with the addition of choline chloride based DES. In another study, more than 40% WS was observed when 40% tea polyphenols (wt%) were added to the chitosan film [45].

4 Conclusion

Total betanins were successfully extracted using ultrasound-assisted extraction by various DES, demonstrating an excellent extraction method. DES extracts were successfully added as a plasticizer to the chitosan biofilm resulting in an enhancement in mechanical properties and antioxidant properties. These changes could result from an interaction between chitosan and DES extract, as evident from FTIR analysis. DES-extract-based chitosan biofilm can present a new perspective for application as a biodegradable packaging film. However, further study is recommended, such as the addition of nanomaterial to reduce the hydrophilic property of the film. Among all the DES-extracted DFP-based chitosan biofilms studied, DES-3 extract film showed the best results as it had high TS, low WS, and WVP values. These films seem to be

an excellent alternative to synthetic film materials and can act as a carrier for active compounds. This can have an impact on food shelf life enhancement and food preservation. Applying DES extract impregnated chitosan film on various food materials can be a follow-up to the reported study.

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Author contribution Rushikesh A. Prajapati: conceptualization, methodology, software, validation, formal analysis, investigation, writing—review and editing. Girirajsinh C. Jadeja: corresponding author, supervision, conceptulization, resources, writing, review and editing, and project administration.

Data availability This is not applicable.

Declarations

Ethical approval This is not applicable. No experiments were conducted on humans or animals as part of this work.

Conflict of interests The authors declare no competing interests.

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